



Finite target-copy MGW synergy

Finite categorical probability identities and scoped formal proofs

Sepehr Mahmoudian

11 September 2026

Result, scope, and evidence status

This note gives an exact finite-law calculation for the categorical shared-exclusions PID of Makkeh, Gutknecht, and Wibral (MGW). The target copies source one and can contain an arbitrary residual coordinate:

$$S_1 \in A, \quad S_2 \in C, \quad U \in B, \quad T = (S_1, U) \in A \times B,$$

where A , C , and B are finite alphabets. No independence or full-support premise is imposed.

The copy direction is from source one into the first coordinate of the target: observing T reveals S_1 exactly, while observing S_1 need not reveal the residual U . This is a premise about the joint law, not a causal claim about how a sensor or target was made. It also does not require the target to copy source two; $U = S_2$ is one special case.

For every positive-mass realization with source values (x, y) , define

$$a = P(S_1 = x), \quad c = P(S_2 = y), \quad j = P(S_1 = x, S_2 = y), \quad d = a + c - j.$$

The local signed-net two-source MGW synergy is

$$s(x, y, u) = \log \frac{ac}{j(a + c - j)} \quad (1)$$

in nats. It does not depend on u or on the conditional law of U after the complete source joint law is fixed. If

$$q(x, y) = P(S_1 = x, S_2 = y), \quad q_1(x) = \sum_y q(x, y), \quad q_2(y) = \sum_x q(x, y),$$

then the averaged synergy is

$$S(P) = \sum_{q(x, y) > 0} q(x, y) \log \frac{q_1(x)q_2(y)}{q(x, y)[q_1(x) + q_2(y) - q(x, y)]} \quad (2)$$

and the exact bounds are

$$0 \leq s(x, y, u) \leq \log \frac{c}{j}, \quad 0 \leq s(x, y, u) \leq \log \frac{a}{j}, \quad (3)$$





$$0 \leq S(P) \leq \min\{H(S_1 | S_2), H(S_2 | S_1)\}. \quad (4)$$

These are law-level identities for the paper-defined categorical MGW functional. The finite-law derivation and its Lean formalization are repository work. This note makes no claim that the identity, bounds, or formalization have novel scientific priority. Source correspondence and priority require their own literature adjudication.

Evidence status. Eleven targets have local Lean acceptance under the stated imports and axiom policy. The [source graph](#) binds their definitions, proofs, and judges; the [execution record](#) preserves those observations. The final table distinguishes these targets from Shannon consequences and examples given only as written derivations. A retrospective audit of the [replay record](#) checked stored outcomes for seventeen source compilations and one same-kernel replay. Complete replay acceptance remains open: the original full-read claim is retracted, and the audited packet lacks outer-return and separate process-ownership transcripts. Private raw records are omitted from the public repository.

Why this setting, and where it comes from

Three objects have different roles here: a probability model, a PID definition, and a derivation. The model specifies finite observations and the deterministic coordinate relation $T = (S_1, U)$. The MGW paper specifies how shared exclusions define information contributions. Equations (1)–(4) are this repository’s scoped derivation for that model and functional. Choosing a copy model does not define a PID, and citing a PID paper does not establish the repository’s proof or its application value.

The two-bit copy benchmark uses independent fair inputs and $T = (S_1, S_2)$. Harder, Salge, and Polani discuss it in their 2012 preprint, Sections II.C and IV.C.1 and Figure 4, before MGW. Their comparison asks whether independent inputs supplying different target coordinates should count as redundant. This documents earlier use; it attributes neither invention to Wibral or this repository nor first priority to Harder and coauthors. [Harder–Salge–Polani, version 3](#).

Here the target may contain only the copied source, both sources, or another residual observation. The ordinary-world construction below assigns a joint mass $w(x, y, u)$ to possible observations and then records the source pair and target. It covers dependent as well as independent worlds; the fair bits used later are transparent examples, not a hidden assumption of the theorem. A physical measurement fits the theorem only if its declared population or empirical law has the stated exact copy relation.

The [recorded office example](#) motivated the added-information question. It uses aligned light and CO₂ values from timestamped records, four training-fitted bins per source, and the original binary occupancy label. In its later recording, positive categorical MGW synergy nearly cancels negative unique CO₂ information. Those are descriptive values for the fitted categorical empirical law; they are not a claim that the occupancy target copies a light bin. The finite copy examples isolate a possible signed cancellation mechanism. They neither model that office nor establish a population sensor ranking.

The finite law and the global copy premise

A complete categorical key is

$$z = ((x, y), (x', u)) \in (A \times C) \times (A \times B).$$





Let $p(z) \geq 0$ and $\sum_z p(z) = 1$. The target-copy premise is global on the nonzero support:

$$p((x, y), (x', u)) > 0 \implies x' = x. \quad (5)$$

It is not enough for (5) to hold only at the key being evaluated. It must hold at every positive-mass key because all event probabilities below sum over the whole law.

This law has an equivalent ordinary-world construction. Given any normalized nonnegative law $w(x, y, u)$ on $A \times C \times B$, define

$$p((x, y), (x', u)) = \begin{cases} w(x, y, u), & x' = x, \\ 0, & x' \neq x. \end{cases} \quad (6)$$

Then

$$\sum_{x, y, x', u} p((x, y), (x', u)) = \sum_{x, y, u} w(x, y, u) = 1,$$

and every nonzero key satisfies (5). The observation map

$$(x, y, u) \mapsto ((x, y), (x, u))$$

is injective, so it preserves each world's mass and turns every finite keyed event into its exact preimage sum. Conversely, a law p satisfying (5) is recovered by $w(x, y, u) = p((x, y), (x, u))$. Thus the theorem covers every finite law for which $T = (S_1, U)$ almost surely, including arbitrary dependence of U on both sources.

For a positive anchor

$$z = ((x, y), (x, u)), \quad \rho = p(z) > 0,$$

write

$$\tau = P(T = (x, u)).$$

All named quantities are actual sums from this one law. In particular,

$$j = \sum_{u' \in B} p((x, y), (x, u')), \quad a = \sum_{y' \in C} q(x, y'), \quad c = \sum_{x' \in A} q(x', y). \quad (7)$$

The positive anchor gives

$$0 < \rho \leq j \leq a, \quad 0 < \rho \leq j \leq c, \quad 0 < \rho \leq \tau. \quad (8)$$

These relations establish the positive logarithm domains used below.

Actual OR/AND events and positive-mass local cumulatives

MGW source-event syntax is an OR across source collections and an AND within a collection. For an antichain α of source-index sets, the event at (x, y) is





$$E_\alpha(x, y) = \bigcup_{J \in \alpha} \bigcap_{i \in J} \{S_i = s_i\}. \quad (9)$$

The four two-source cumulative nodes therefore use

$$E_1 = \{S_1 = x\}, \quad E_2 = \{S_2 = y\}, \quad E_{12} = E_1 \cap E_2, \quad E_{\text{red}} = E_1 \cup E_2. \quad (10)$$

The joint-source node is the within-collection AND. The redundancy node is the across-collection OR. Inclusion-exclusion gives the actual redundancy-event mass

$$P(E_{\text{red}}) = P(E_1) + P(E_2) - P(E_1 \cap E_2) = a + c - j = d. \quad (11)$$

Let $F = \{T = (x, u)\}$ be the actual target event. On positive support, F already implies $S_1 = x$ by (5). Consequently $E_1 \cap F = F$ and $E_{\text{red}} \cap F = F$ up to zero-mass keys. The event $E_2 \cap F$ fixes $S_2 = y$, while F and (5) fix $S_1 = x$ and the complete target (x, u) , so it selects the anchor itself. The same is true of $E_{12} \cap F$. Hence the exact event masses are

cumulative node	source event	$P(E)$	$P(E \cap F)$
source one	E_1	a	τ
source two	E_2	c	ρ
joint sources	E_{12}	j	ρ
redundancy	E_{red}	$d = a + c - j$	τ

Every entry in the last two columns is positive: the relevant event contains the positive anchor. Also $d \leq 1$ because it is a probability.

For any one of these source events, the MGW informative and misinformative local cumulatives in nats are

$$K^+(E) = -\log P(E), \quad K^-(E; F) = \log \frac{P(F)}{P(E \cap F)} = -\log P(E | F), \quad (12)$$

and the signed net cumulative is

$$K^{\text{net}}(E; F) = K^+(E) - K^-(E; F) = \log \frac{P(E \cap F)}{P(E)P(F)}. \quad (13)$$

The event masses are positive, so no totalized value such as $\log 0$ enters the local information. Both K^+ and K^- are nonnegative because $P(E) \leq 1$ and $0 < P(E \cap F) \leq P(F)$. Their signed difference need not be nonnegative. Substitution gives

node	K^+	K^-	K^{net}
source one	$-\log a$	0	$-\log a$
source two	$-\log c$	$\log(\tau/\rho)$	$\log[\rho/(c\tau)]$
joint sources	$-\log j$	$\log(\tau/\rho)$	$\log[\rho/(j\tau)]$
redundancy	$-\log d$	0	$-\log d$

This table is the probability step that prevents the symbols a, c, j, d, ρ, τ from becoming unconstrained algebra parameters.





Local identity

The two-source Möbius row for the signed-net synergy atom is joint minus source one minus source two plus redundancy. Using the last column of the table,

$$\begin{aligned}
 s(z) &= K_{12}^{\text{net}} - K_1^{\text{net}} - K_2^{\text{net}} + K_{\text{red}}^{\text{net}} \\
 &= \log \frac{\rho}{j\tau} - (-\log a) - \log \frac{\rho}{c\tau} - \log d \\
 &= (\log \rho - \log j - \log \tau) + \log a - (\log \rho - \log c - \log \tau) - \log d \\
 &= \log a + \log c - \log j - \log d \\
 &= \log \frac{ac}{jd} = \log \frac{ac}{j(a+c-j)}.
 \end{aligned} \tag{14}$$

All logarithm rules in (14) apply to positive quantities by (8) and the positive event-mass argument. Both the anchor mass ρ and target-event mass τ cancel. This proves (1) and shows locally why the residual target coordinate drops out.

Averaging, marginalization, and invariance

The averaged pointwise atom is the positive-support sum

$$S(P) = \sum_{z:p(z)>0} p(z)s(z). \tag{15}$$

Equation (14) is constant over all positive target residuals at a fixed source row (x, y) . Since p is nonnegative and the alphabets are finite,

$$\sum_{u:p((x,y),(x,u))>0} p((x,y),(x,u)) = q(x,y). \tag{16}$$

Moreover, $q(x, y) > 0$ exactly when at least one such residual-target cell has positive mass. Regrouping (15) by source row and using $a = q_1(x)$, $c = q_2(y)$, and $j = q(x, y)$ gives (2) exactly.

Therefore two target-copy laws with the same complete source joint law q have the same averaged signed-net synergy. Equality of the separate marginals q_1 and q_2 is insufficient because both $j = q(x, y)$ and $d = q_1(x) + q_2(y) - q(x, y)$ retain source dependence.

The current formal comparison quantifies two laws over one common residual alphabet B . It does not export cross-type invariance between unrelated residual alphabets. Unused values of that common B are allowed, so constant and nonconstant residual channels can still be compared without changing the formal target type.

There is also a useful written corollary, not a separate Lean export in this package. The averaged redundancy in this family is

$$R = - \sum_{q(x,y)>0} q(x,y) \log d(x,y),$$





while

$$I(S_1; S_2) = \sum_{q(x,y)>0} q(x,y) \log \frac{q(x,y)}{q_1(x)q_2(y)}.$$

Termwise subtraction yields

$$S = R - I(S_1; S_2). \quad (17)$$

This is another statement about this exact target-copy family, not a general identity for MGW synergy under arbitrary target channels.

Local and averaged bounds

For a positive source cell, $j \leq a$ and $j \leq c$. Thus $a > 0$, $c > 0$, and

$$d = a + c - j \geq a > 0, \quad d \geq c > 0. \quad (18)$$

The lower bound follows from one exact factorization:

$$ac - jd = ac - j(a + c - j) = ac - aj - cj + j^2 = (a - j)(c - j) \geq 0. \quad (19)$$

Because $jd > 0$, equation (19) gives $ac/(jd) \geq 1$. The logarithm is increasing, so $s \geq \log 1 = 0$.

For the first upper bound, use $a/d \leq 1$ from $d \geq a$:

$$\frac{ac}{jd} = \frac{c}{j} \frac{a}{d} \leq \frac{c}{j}, \quad s \leq \log \frac{c}{j}. \quad (20)$$

For the other bound, use $c/d \leq 1$ from $d \geq c$:

$$\frac{ac}{jd} = \frac{a}{j} \frac{c}{d} \leq \frac{a}{j}, \quad s \leq \log \frac{a}{j}. \quad (21)$$

The right sides are the two local conditional surprisals:

$$\log \frac{c}{j} = -\log P(S_1 = x \mid S_2 = y), \quad \log \frac{a}{j} = -\log P(S_2 = y \mid S_1 = x). \quad (22)$$

Multiply each resulting logarithmic bound, $0 \leq s$, (20), and (21), by $q(x, y) \geq 0$ and sum only over $q(x, y) > 0$. The lower bounds sum to $S \geq 0$, and the upper bounds give

$$S \leq \sum_{q(x,y)>0} q(x,y) \log \frac{q_2(y)}{q(x,y)} = H(S_1 \mid S_2), \quad (23)$$

$$S \leq \sum_{q(x,y)>0} q(x,y) \log \frac{q_1(x)}{q(x,y)} = H(S_2 \mid S_1). \quad (24)$$

Equations (23) and (24) prove (4). The right sides are the usual finite conditional entropies: averages of conditional surprisal under the actual joint law. This convention follows [Shannon's conditional-entropy definition](#), Section 6, item 5. The Lean interface defines these positive-support sums explicitly; this package does not claim a separate identification theorem with another entropy library. A positive $q(x, y)$ makes both relevant marginals positive, so the sums have no zero denominator.





Three exact examples

These are written specializations and consequences of the formulas above. Their arithmetic is shown explicitly; they are not separate named exports in the eleven-target Lean roster.

1. Fair independent sources and an arbitrary residual target

Let $A = C = B = \{0, 1\}$, and let S_1 and S_2 be independent fair bits. Keep any conditional kernel $P(U = u \mid S_1 = x, S_2 = y)$; it can depend arbitrarily on both sources. For every source pair,

$$a = c = \frac{1}{2}, \quad j = \frac{1}{4}, \quad d = \frac{3}{4}.$$

Therefore every positive target-copy key has

$$s = \log \frac{(1/2)(1/2)}{(1/4)(3/4)} = \log \frac{4}{3},$$

and the weights in (15) sum to one, so

$$S = \log \frac{4}{3} \tag{25}$$

for every residual channel. This already proves that synergy cannot rank residual-target task gain while the source joint law stays fixed. Two channels over the same formal alphabet B make the point exact:

- If $U = 0$ almost surely, then $T = (S_1, 0)$ and $I(S_2; T \mid S_1) = 0$.
- If $U = S_2$, then $T = (S_1, S_2)$ and $I(S_2; T \mid S_1) = H(S_2 \mid S_1) = \log 2$.

Both channels have the same synergy (25). They have different information increments. This comparison changes the residual target channel and thus changes the prediction task. Ordinary sensor selection instead fixes a world and target while comparing available sources. The **fixed-world example** makes that latter comparison: target (A, B) , baseline A , and additions consisting of either an independent bit or B . Both additions have the same categorical MGW synergy, but only B adds target information. The two examples address distinct experimental designs.

Comparison	Held fixed	Changed	Question isolated
Residual-target example above	The complete source law q	$U = 0$ versus $U = S_2$ in $T = (S_1, U)$	Can changing target information leave this synergy unchanged?
Fixed-world companion	The world, target (A, B) , and baseline source A	Add the independent bit U or the missing target bit B	Can different candidate sources tie in synergy while adding different information?

The companion's A, B, U are mutually independent fair bits, with U naming the unrelated candidate. They are its own variable names, not an independence assumption on the arbitrary residual U in the general theorem. Writing U_C for the chosen candidate's unique atom, the already derived pairs (U_C, S) are $(-\log(4/3), \log(4/3))$ and $(\log 2 - \log(4/3), \log(4/3))$. These signed sums distinguish zero added information from $\log 2$ nats while the target remains fixed.

The signed two-source reconstruction explains the cancellation. If U_2 is source two's unique atom and R is redundancy, then





$$I(S_2; T | S_1) = U_2 + S. \quad (26)$$

Here $R = \log(4/3)$. For constant U , $I(S_2; T) = 0$, hence $U_2 = -\log(4/3)$ and $U_2 + S = 0$. For $U = S_2$, $U_2 = \log 2 - \log(4/3)$ and $U_2 + S = \log 2$. The negative unique atom in the first channel is a valid signed MGW contribution. Clamping it would break (26).

2. Copied sources give zero synergy

Let the sources share one finite alphabet and suppose $S_2 = S_1$ almost surely. For every supported value x , put $\mu_x = P(S_1 = x) > 0$. Then

$$a = c = j = d = \mu_x,$$

so

$$s = \log \frac{\mu_x^2}{\mu_x^2} = 0, \quad S = 0. \quad (27)$$

For fair copied bits, the only positive source cells are $(0, 0)$ and $(1, 1)$, each with mass $1/2$; the two off-diagonal zero cells are simply absent from the support sum. The conclusion is unchanged for any residual channel compatible with the copied sources.

3. A dependent 2×2 source law

Let $A = C = \{0, 1\}$, take a singleton residual alphabet, and use the source law

$$\begin{array}{c|cc} q(x, y) & y = 0 & y = 1 \\ \hline x = 0 & 3/8 & 1/8 \\ x = 1 & 1/8 & 3/8 \end{array} \quad (28)$$

with $T = (S_1, *)$. Both marginals are fair. On a diagonal cell,

$$j = \frac{3}{8}, \quad d = \frac{5}{8}, \quad s_{\text{diag}} = \log \frac{1/4}{(3/8)(5/8)} = \log \frac{16}{15}.$$

On an off-diagonal cell,

$$j = \frac{1}{8}, \quad d = \frac{7}{8}, \quad s_{\text{off}} = \log \frac{1/4}{(1/8)(7/8)} = \log \frac{16}{7}.$$

The two diagonal cells have total weight $3/4$, and the two off-diagonal cells have total weight $1/4$. Thus

$$S = \frac{3}{4} \log \frac{16}{15} + \frac{1}{4} \log \frac{16}{7}. \quad (29)$$

This differs from the independent fair-source value despite identical separate marginals. In fact it is strictly smaller, without using a decimal approximation. Multiplying by four and exponentiating reduces the comparison to

$$\frac{16^4}{15^3 7} < \frac{4^4}{3^4},$$





which is equivalent to

$$4^4 \cdot 3^4 = 20736 < 23625 = 15^3 \cdot 7.$$

The example isolates why the invariance theorem requires the exact joint source law q , not only q_1 and q_2 .

The source joint law determines target-copy synergy

Categorical MGW · target $T = (S_1, U)$ · arbitrary finite law · nats

At an anchor (x, y) : $a = P(S_1 = x)$, $c = P(S_2 = y)$, $j = P(S_1 = x, S_2 = y)$.

The union $D = \{S_1 = x\} \text{ OR } \{S_2 = y\}$ has probability $d = a + c - j$.

Diagonal anchor (0, 0)

	$S_2 = 0$	$S_2 = 1$
$S_1 = 0$	3/8	1/8
$S_1 = 1$	1/8	3/8

$$a = c = 1/2; j = 3/8; d = 5/8$$

$$s = \ln[(1/4) / ((3/8)(5/8))]$$

$$= \ln(16/15)$$

Off-diagonal anchor (0, 1)

	$S_2 = 0$	$S_2 = 1$
$S_1 = 0$	3/8	1/8
$S_1 = 1$	1/8	3/8

$$a = c = 1/2; j = 1/8; d = 7/8$$

$$s = \ln[(1/4) / ((1/8)(7/8))]$$

$$= \ln(16/7)$$

 Union D: shaded cells

 Anchor: inner frame

Numbers give cell masses.

Both diagonal cells weigh 3/4 in total; both off-diagonal cells weigh 1/4.

Joint-law average: $S = (3/4) \ln(16/15) + (1/4) \ln(16/7)$.

Why the residual target channel disappears

At a positive full key, let $\rho = P(S_1 = x, S_2 = y, U = u)$ and $\tau = P(T = (x, u))$.

$$s = \ln[\rho/(j\tau)] + \ln a - \ln[\rho/(c\tau)] - \ln d = \ln[ac/(jd)].$$

The positive masses ρ and τ cancel. The full joint source law remains essential.

Figure 1: The complete source joint law determines target-copy synergy. Shading shows the OR event, and the inner frame marks the anchor cell. The two grids use the same joint law; changing the anchor changes its intersection mass and union mass. The lower calculation shows why positive target masses cancel.

Comparing definitions on the copy benchmark

A common author or the word “synergy” does not identify a single PID functional. The categorical MGW construction and the continuous construction of Ehrlich and coauthors, including Wibral, have different definitions and support requirements. Other PIDs make different choices again. The already specified independent two-bit copy benchmark makes the difference concrete. At the all-zero anchor, $T = (S_1, S_2)$ has four equally likely values. The source-match OR retains (0, 0), (0, 1), and (1, 0). Knowing that event raises the target-anchor probability from 1/4 to 1/3, so MGW redundancy





is $\log(4/3)$ despite source independence. Harder’s identity criterion instead requires redundancy for a copied pair to equal the sources’ mutual information, which is zero here. The constructions ask different questions about what should count as shared information. [Harder, Section III.A](#).

The table compares definitions on this benchmark, not estimator accuracy or application rankings.

Definition	What the comparison establishes
Categorical MGW shared exclusions	The local source event is the inclusive OR of matching source values. For independent fair sources and their copied pair as target, example 1 gives $S = \log(4/3)$. This is one signed atom under that event definition. MGW, Section II, equations (1)–(3).
Williams–Beer $I_{\min}$	Redundancy averages the minimum, over sources, of the information specific to each target outcome. It is not the minimum of two averaged MIs. Harder’s published copy comparison assigns it positive redundancy even for independent inputs. Williams–Beer, Section III, equations (2)–(3); Harder, Table I.
Harder–Salge–Polani I_{red}	For each outcome of one source, the target’s conditional distribution is projected onto convex combinations of target distributions conditioned on outcomes of the other source. The redundancy takes the smaller projected-information value from the two source directions. Its independent-copy example consists of unique information from each input, with zero redundancy and synergy. This matches its additional identity requirement. Section III.B, equations (6), (8), and (11)–(13); Section III.A; Figure 4.
Bertschinger–Rauh–Olbrich–Jost–Ay (BROJA)	Unique information is defined by minimizing CMI over laws with the same two source–target marginals. Its published identity result assigns zero synergy when the target is the source pair. Section 3 and Proposition 18.
Ehrlich and coauthors’ continuous shared exclusions	Its density-sum functional and nearest-neighbour estimator have separate support and relative-scale assumptions. They are not evaluated by substituting the finite copy masses into a continuous formula. Definition 2 and Appendix C.

These are different axiomatic choices. In categorical MGW, a net atom is the informative atom minus the misinformative atom; nonnegativity of the two components does not imply nonnegativity of their difference. Harder’s additional copy-identity axiom and a requirement of nonnegative net atoms are not assumptions of the MGW derivations in this note. Each comparison must state which requirement is being tested. MGW, [Section IV.A](#).

The finite-law comparisons keep the same Shannon information available from the observations while assigning it to different atoms. A study that requires independent target coordinates to have zero redundancy should state that requirement before choosing its PID. A study that asks about the information carried by a particular source-match disjunction may instead use MGW’s local event construction. Neither choice alone supplies a sensor-selection rule. The continuous row identifies a separate object; no categorical-to-continuous equivalence or comparative numerical result is claimed here.

The CMI complement and practical use

For this categorical MGW two-source decomposition, the reconstruction identities are

$$\begin{aligned}
 I(S_1; T) &= R + U_1, \\
 I(S_2; T) &= R + U_2, \\
 I(S_1, S_2; T) &= R + U_1 + U_2 + S.
 \end{aligned}$$





For finite variables, define $I(X; Y) = H(Y) - H(Y | X)$ and $I(X; Y | Z) = H(Y | Z) - H(Y | X, Z)$. Subtracting the two MI expressions cancels $H(Y)$, which gives the conditional-MI identity below. Subtracting the first MGW reconstruction line from the third then gives (26):

$$I(S_2; T | S_1) = I(S_1, S_2; T) - I(S_1; T) = U_2 + S. \quad (30)$$

In the target-copy family, this increment is also

$$I(S_2; T | S_1) = I(S_2; U | S_1). \quad (31)$$

Equations (26), (30), and (31) are written Shannon/reconstruction consequences, not additional formal exports. In (31), conditioning on S_1 fixes the copied coordinate of T ; only U remains uncertain, before and after observing S_2 .

The synergy atom separates one definition-specific part of the increment; it is not the increment itself. A positive S can be exactly cancelled by negative U_2 . This is the correct signed interpretation and is the main safeguard for analysis, training, and sensor selection.

MGW's local description asks how learning the source-match OR changes the probability of the observed target. Its informative and misinformative components retain contributions from the exclusions. An averaged CMI does not display that event description. MGW, Sections II, IV.A and VI.C.

Decomposing each joint outcome lets an analyst inspect the observations behind an average, as the four-state OR calculation illustrates. Separate informative and misinformative components retain exclusion contributions that their net difference can hide. On the interior of a finite joint-probability simplex, MGW's differentiable population quantities provide a basis for gradient-based objectives. These capabilities do not establish estimator-gradient accuracy or training gains. The exact-copy qualification below and the boundary cases still apply. MGW, Sections IV.A–B, Section V.E.1.

A negative net unique atom remains an algebraic contribution in this specific decomposition; it does not diagnose a defective sensor, deceptive intent, or a harmful causal effect.

If the question is simply how much additional Shannon information a candidate supplies after a baseline, CMI is the direct quantity and no PID is needed. If the question is whether a particular system makes better decisions, compare its held-out task loss, with a stated target, prediction rule, validation design, and acquisition cost. The office example already reports such a bounded comparison: adding CO₂ slightly improves later-recording classification accuracy but worsens its probability scores. No PID-trained predictor was evaluated there. These outcomes distinguish information, decomposition, and achieved predictive performance.

For analysis, equations (1) and (2) provide an exact reference calculation for a declared finite law. They can expose which source cells produce large signed-net synergy and can test a future implementation after a separate code-to-formal-object mapping is established. Equation (17) clarifies the mechanism in this family: source dependence and the shared-exclusion union mass fully determine the result.

For training, include the fair-source example as a negative control when the proposed objective fits this target-copy setting. If a learner controls a residual target representation U , it can change that channel from constant to fully revealing while S remains $\log(4/3)$ and the source law stays fixed. This comparison changes the represented prediction task. It does not establish failure for every learning method with a fixed target. When the objective is added predictive information, compare CMI or the complete signed sum $U_2 + S$; for an actual trained predictor, also measure held-out task loss. A different atom weighting defines a different objective and needs its own benefit evidence.

Differentiability also needs a domain. MGW's Section IV.B establishes it on the interior of the finite joint-probability simplex. With a nontrivial copied-source alphabet and the unrestricted





source–target product alphabet, (5) forces structural zero mass on incompatible coordinates. The copy law therefore lies on the boundary of that ambient simplex. The interior result cannot be transferred to it without another argument. Neither smoothness along a chosen fixed-support parameterization nor useful gradients for a trained estimator are proved here. Adding noise or positive mass changes the law and may remove the exact-copy premise. MGW, Section IV.B.

For sensor selection, the formula can diagnose a proposed source grouping, but it does not include acquisition cost, failure cost, latency, coverage, causal effect, or learned-predictor error. A selection claim must compare the relevant CMI or task-risk change, fixed-model and retrained ablations, and the chosen decision cost. Positive synergy alone does not establish that adding a sensor improves the task.

Boundary cases and nonclaims

Zero probability cells are allowed. Local quantities are asserted only at $p(z) > 0$, and averaged quantities sum only over positive support. Thus the result never assigns scientific meaning to totalized division by zero or $\log 0$. Singleton alphabets are also allowed.

If the complete key space is empty, the sum of every nonnegative law over it is zero, so no normalized law exists. The all-zero mass function likewise fails normalization. The theorem is therefore vacuous for an empty law rather than an extension that assigns it information values.

An exact finite PMF needs no IID premise for these probability identities. If rows are used to form an empirical PMF, the result describes that empirical law exactly. It does not turn the rows into IID observations, prove that an observed copy relation holds in the population, control unseen or rare cells, establish convergence or confidence coverage, or calibrate a finite-sample estimator. Those claims require a sampling model and their own theorems.

The result also does not establish Rust or Python refinement, binary64 logarithm error, resource bounds, continuous-variable shared exclusions, another PID functional, approximate-copy stability, cross-residual-alphabet formal invariance, or downstream performance. It does not close the three-source program or the wider pid-rs program.

Formal target map

The frozen Lean interface divides the argument into eleven unchanged propositions:

targets	mathematical role
<code>WorldBridgeTarget</code>	Push a normalized finite world law through the injective target-copy observation and preserve event preimages.
<code>CopyEventMassesTarget</code>	Prove the four target-restricted masses τ, ρ, ρ, τ from global support copying.
<code>SourceUnionMassTarget</code>	Prove $d + j = a + c$ by exact finite inclusion-exclusion.
<code>LocalCumulativesTarget</code>	Derive the four signed-net cumulative formulas from actual positive event masses.
<code>SourceProjectionTarget</code>	Identify a, c, j with the true source marginal and regroup every source-row integrand.
<code>LocalIdentityTarget,</code> <code>WorldLocalIdentityTarget</code>	Establish (1) for the imported MGW signed-net synergy at a positive key and at an observed world.
<code>AveragedIdentityTarget,</code> <code>SameSourceLawTarget</code>	Establish (2) and same- q invariance for two laws over one common B .





targets	mathematical role
LocalBoundsTarget, AveragedBoundsTarget	Establish (3) and (4).

A checked theorem proves only its exact proposition under its imported definitions, axioms, and toolchain. It does not by itself prove publication-to-Lean correspondence, executable refinement, statistical calibration, application value, independent replay, or publication on the project mainline.

References

Abdullah Makkeh, Aaron J. Gutknecht, and Michael Wibral, “Introducing a Differentiable Measure of Pointwise Shared Information,” *Physical Review E* **103**, 032149 (2021), doi:10.1103/PhysRevE.103.032149, arXiv:2002.03356v5. The paper defines the categorical shared-exclusions construction, including the OR-of-AND source event, the informative/misinformative split, the pointwise functional, the lower-cumulative atom relation, and joint-law averaging. This note uses natural logarithms and therefore reports nats; bit-valued expressions scale by the positive factor $\log 2$.

Malte Harder, Christoph Salge, and Daniel Polani, “A Bivariate Measure of Redundant Information,” arXiv:1207.2080v3 (2012). Sections II.C, III.A, and IV.C.1 distinguish the copy benchmark, the identity criterion, and their projection-based redundancy. Figure 4 and Table I supply the comparisons used here.

Paul L. Williams and Randall D. Beer, “Nonnegative Decomposition of Multivariate Information,” arXiv:1004.2515v1 (2010), Section III, equations (2)–(3). This is the defining source for $I_{\min}$, a different redundancy functional.

Nils Bertschinger, Johannes Rauh, Eckehard Olbrich, Jürgen Jost, and Nihat Ay, “Quantifying Unique Information,” *Entropy* **16**, 2161–2183 (2014), doi:10.3390/e16042161, arXiv:1311.2852v2, Section 3 and Proposition 18.

David A. Ehrlich and coauthors, “Partial Information Decomposition for Continuous Variables Based on Shared Exclusions: Analytical Formulation and Estimation,” *Physical Review E* **110**, 014115 (2024), doi:10.1103/PhysRevE.110.014115, arXiv:2311.06373v3, Definition 2 and Appendix C. This is a separate continuous functional and estimator; the finite target-copy identities do not establish its support admission, estimator calibration, or application value.

Claude E. Shannon, “A Mathematical Theory of Communication,” *Bell System Technical Journal* **27**, 379–423 and 623–656 (1948), Section 6, item 5 and its entropy addition rule. Reprint. This is the classical entropy source, not the definition of categorical MGW PID.

